

Nonthermal nonlinear many-body excitonic dynamics in high-purity silicon

Patrick Pilch,¹ Siyu Duan,¹ Yang Liu,^{2,3} Manfred Bayer,^{1,4,5} Marc Assmann,^{1,5} and Zhe Wang^{1,5}

¹*Department of Physics, TU Dortmund University, 44227 Dortmund, Germany*

²*Center for Correlated Matter and School of Physics, Zhejiang University, Hangzhou, China*

³*Institute for Advanced Study in Physics, Zhejiang University, Hangzhou 310058, China*

⁴*Research Center FEMS, TU Dortmund University, 44227 Dortmund, Germany*

⁵*Dortmund Center for Advanced Exploration of Dynamics Across Limits Using Spectroscopy (DAEDALUS), TU Dortmund University, 44227 Dortmund, Germany*

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We report time-resolved optical pump-terahertz third-harmonic generation spectroscopy of picosecond non-thermal many-body charge dynamics in high-purity silicon at room temperature. Although in thermal equilibrium excitons in silicon are unstable due to a lower binding energy than the thermal energy at room temperature, following pulsed non-resonant optical pumping we can identify the formation of transient excitonic states on the picosecond time scale in our high-resolution nonlinear spectroscopic measurements. In contrast to unbound electron-hole pairs, the nonthermal many-body excitonic dynamics feature a characteristic time-dependent evolution, a distinct dependence on charge density, and a nonperturbative nonlinear terahertz response.

An exciton is a bosonic composite quasiparticle of an electron-hole pair bound by attractive Coulomb interaction. The bosonic character and many-body interactions of excitons make them fundamentally appealing for the realization of exotic quantum phenomena in solid-state materials, e.g. Bose-Einstein condensation [1, 2], exciton fluids [3], excitonic insulators [4, 5] or exciton crystals [6]. The formation and transport dynamics of excitons are crucial also for applications, such as semiconductor optoelectronics or quantum technologies [7, 8]. Experimental studies of excitonic dynamics have mainly been based on spectroscopic methods focusing on the linear-response regime. For example, photoluminescence spectroscopy measures emitted photons due to recombination of the electron and hole in an exciton [1–3]; Absorption or reflection spectroscopy detects the coherent excitonic polarization induced by optical excitation [2, 6, 9]; Also incoherent excitons can be probed by their electromagnetic field-driven transport properties particularly in the terahertz (1 THz = 10^{12} Hz) frequency range [10–13], while a *dc* field cannot induce a current of the charge-neutral objects.

In this work, by using a different spectroscopic technique that goes beyond the linear response and thermal equilibrium – time-resolved optical pump THz high-harmonic generation spectroscopy – we investigate many-body excitonic dynamics in the indirect bandgap semiconductor silicon at room temperature. Whereas without optical excitation the high-purity silicon is transparent for THz electromagnetic waves, with optical excitation of electron-hole pairs and strong THz drive pulses we are able to detect THz third-harmonic generation (THG), which corresponds to the lowest order nonzero nonlinearity since second-order nonlinear responses vanish due to inversion symmetry of silicon. In this optically-excited strongly interacting many-body system, complex dynamical processes are taking place simultaneously. The observation of the high-order nonlinear response allows us to sensitively track the formation or dissociation of excitons down to picosecond time scales. In particular, we resolve characteristic THz third-harmonic generation of the far-from-equilibrium

picosecond excitonic dynamics by their distinctive dependence on charge carrier density featuring a Mott crossover and their nonperturbative dependence on THz drive strength.

Charge dynamics in silicon plays a key role in its widespread applications, ranging from computers and smart phones, to lasers, LEDs, and solar cells. Characterized by an indirect band gap of about 1.12 eV at room temperature, phonon-assisted excitation of electron-hole pairs in silicon can be triggered by visible or near-infrared photons. The binding energy of excitons in silicon is about 14.7 meV [14], corresponding to a thermal energy of $T = 170$ K. Therefore, in thermal equilibrium, excitons in silicon are not stable at room temperature due to strong thermal fluctuations. Hence, previous THz spectroscopic studies of excitonic dynamics in silicon were performed at liquid-helium temperatures where excitons are thermally stable and a coherent intraexcitonic absorption line of the quasi-equilibrium excitons is well resolvable [15–17]. In contrast to these previous studies, here we study a very different regime of excitonic dynamics, i.e. far-from-equilibrium picosecond ultrafast nonlinear dynamics, which is distinct on the following aspects: Firstly, we focus on room temperature where the lifetime of excitons is expected to be shorter due to enhanced thermal fluctuations compared with cryogenic temperatures. Secondly, following the optical creation of electron-hole pairs, with strong THz pulse we drive the nonthermal many-body system even further away from thermal equilibrium. Thirdly, we employ the time-resolved nonlinear (rather than linear) response to sense many-body dynamics of the strongly interacting system.

For our spectroscopic study, we use an intrinsic high-resistive $\langle 100 \rangle$ -oriented silicon wafer with resistivity $> 10^5 \Omega\text{cm}$ and a thickness of 0.5 mm. To excite electron-hole pairs across the indirect gap, we use 1.55 eV photons [15–17, 19, 20] from a pulsed laser system (800 nm) with a pulse duration of 100 fs and a repetition rate of 1 kHz. Since silicon has a direct band gap of 3.43 eV, electron-hole pairs across this gap cannot be directly excited by 1.55 eV photons through either linear absorption or second-order nonlinear op-

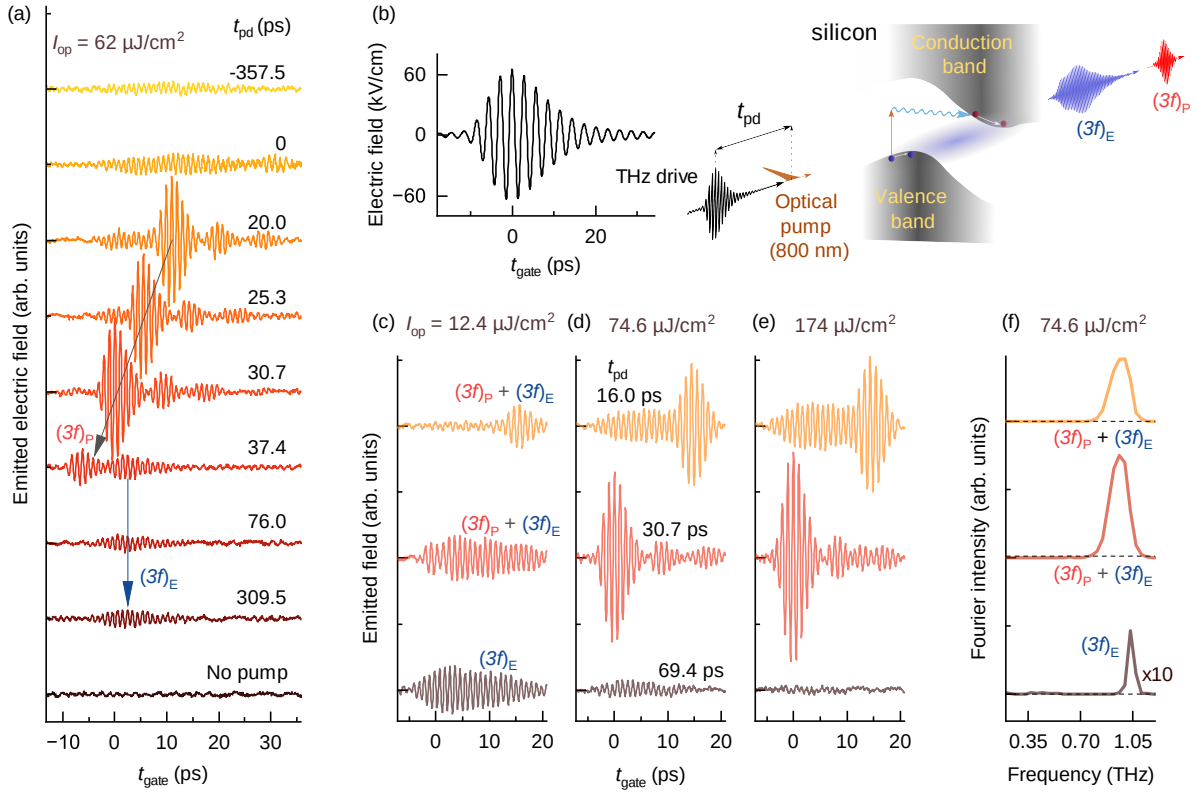


Figure 1. (a) THz third harmonic amplitudes for various pump-drive delays t_{pd} at a fixed optical pump fluence of $I_{\text{op}} = 62 \mu\text{J}/\text{cm}^2$ and a driving THz electric field of 65 kV/cm. With varying t_{pd} , a shifting and a non-shifting third-harmonic signal are resolved as indicated by the arrows and denoted by $(3f)_P$ and $(3f)_E$ for plasma and excitonic emission, respectively. (b) Waveform of the 0.35 THz drive pulse, and schematics of THz field-driven charge carrier dynamics after phonon-assisted optical excitations in the indirect bandgap semiconductor silicon. The valence and conduction bands are characterized by nonparabolic dispersion relations, which is crucial for efficient nonlinear THz emission [18]. Emitted third-harmonic electric field in the time domain measured at three well-separated pump-drive delays t_{pd} for optical pump fluences of (c) $I_{\text{op}} = 12.4$, (d) 74.6 , and (e) $174 \mu\text{J}/\text{cm}^2$. (f) The corresponding frequency-domain spectra for $I_{\text{op}} = 74.6 \mu\text{J}/\text{cm}^2$.

tical processes. Based on the same laser system, our intense multicycle THz drive pulses are obtained by bandpass filtering THz radiation from optical rectification in a LiNbO_3 crystal [21, 22]. For an $f = 0.35$ THz drive pulse [see Fig. 1(b) for its waveform], we used two 0.35 THz bandpass filters each with $\pm 26\%$ bandwidth at -20 dB. With respect to the THz drive, the arrival time of the optical pump pulse at the sample position is tunable, with the corresponding pump-drive time difference denoted by t_{pd} [see Fig. 1(b) for illustration]. t_{pd} is defined as positive when the pump arrives before the drive pulse. The THz electric field is determined by electro-optic sampling with a ZnTe crystal [23], in which the time delay of a gating optical pulse with respect to the THz pulse is denoted by t_{gate} . The measurements are performed at room temperature under ambient conditions.

Under the drive of an $f = 0.35$ THz pulse with a peak electric field of 65 kV/cm, the emitted third-harmonic radiation is recorded after two 1 THz bandpass filters each with $\pm 30\%$ bandwidth at -20 dB. The obtained time-domain traces are presented in Fig. 1(a) for various pump-drive delays and an optical pump fluence of $62 \mu\text{J}/\text{cm}^2$. For contrast, the trace recorded without optical pump displays no THz

third-harmonic oscillations, which confirms the high purity of the intrinsic silicon sample, since doped silicon samples are not transparent in the THz frequencies and emit THz third-harmonic radiation even without optical pump [24–26]. A strong THG yield appears in intrinsic silicon after the optical excitation, reflecting THz driven nonlinear charge dynamics of the optically excited states. As indicated by the two arrows in Fig. 1(a), two third-harmonic traces denoted by $(3f)_P$ and $(3f)_E$, respectively, can be distinguished by their different dependencies on the pump-drive delay t_{pd} . While the $(3f)_P$ traces shift in the time domain following the shifting optical pump with respect to the THz drive pulse, the $(3f)_E$ traces remain at the same position. Besides their very different dependence on the pump-drive delay, as will be discussed below, the two traces exhibit a rich set of distinctly different spectral characteristics. These features allow us to assign the $(3f)_P$ and $(3f)_E$ traces as plasma and excitonic THG, respectively.

It is worth emphasizing that the excitonic third-harmonic emission can be distinguished from the plasma THG with a temporal separation even shorter than 8 ps [see e.g. the $t_{\text{pd}} = 37.4$ ps curve in Fig. 1(a)], whereas from a linear-response probe [15] a coherent intraexcitonic $1S$ - $2P$ excita-

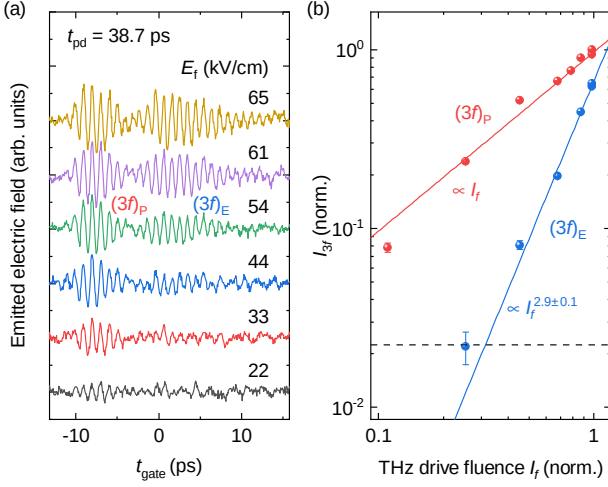


Figure 2. Time-domain third-harmonic traces measured with $I_{op} = 74.6 \mu\text{J}/\text{cm}^2$ for different THz drive fields from 22 to 65 kV/cm (a) at $t_{pd} = 38.7$ ps where two third-harmonic contributions $(3f)_E$ and $(3f)_P$ are well separated. (b) The integrated third-harmonic intensity I_{3f} of the plasma contribution exhibits a linear dependence on the THz drive fluence I_f , while for the exciton contribution $I_{3f} \propto I_f^{2.9 \pm 0.1}$. The noise level is indicated by the horizontal dashed lines.

tion became resolvable at least 50 ps after the optical excitation. This indicates that excitonic correlations are built up from the electron-hole plasma on a shorter time scale due to the Coulomb interaction which is sensitively probed in our nonlinear spectroscopy, while a fully bound electron-hole pair is formed at later stage which is manifested by the resonant absorption in linear THz probe spectroscopy [27]. The rapid building up of excitonic correlations relative to the THz drive-pulse duration of approximately 30 ps [Fig. 1(b)] and the relatively long lifetime of excitons [15] allow the detection of the temporally separated plasma and excitonic THG for a single pump-drive delay at $t_{pd} \gtrsim 30$ ps [e.g. $t_{pd} = 37.4$ ps in Fig. 1(a)]. Moreover, no excitonic second-harmonic generation was measurable under the same conditions, which is consistent with the centrosymmetric structure of silicon and excludes substantial nonlinear effects due to symmetry breaking at the surfaces or field-induced symmetry breaking.

In addition, we observe a weak third-harmonic signal at negative pump-drive delays, see e.g. $t_{pd} = -357.5$ ps in Fig. 1(a), in contrast to the intuition that any THz third-harmonic yield should appear after the optical pump. This signal corresponds to the dynamics of the residual photoexcited electrons and holes from the preceding pulse, which is one millisecond earlier. Since under quasi-thermal equilibrium conditions fully bound excitons cannot survive on millisecond time scales at room temperature, the observed slow dynamics may be most intuitively attributed to trapped carriers due to possible imperfection at the sample surface. Dynamics on such slow time scales in this indirect bandgap semiconductor [28] has been subject of previous studies by using time-resolved linear THz spectroscopy [19, 20, 29]. In con-

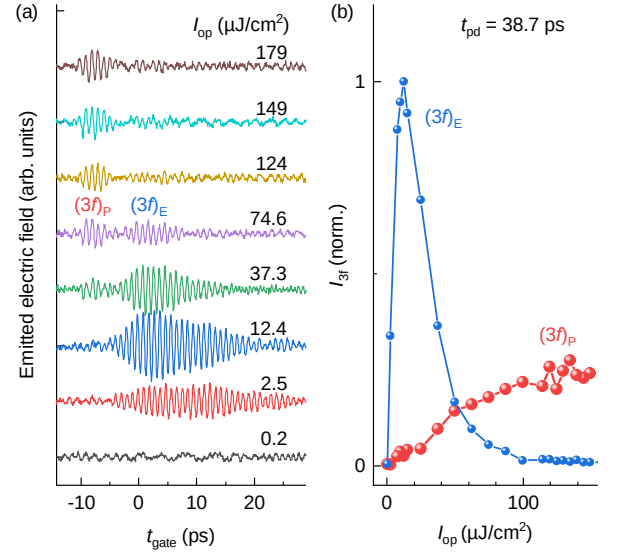


Figure 3. (a) Optical pump fluence dependence of third-harmonic yields measured with a THz drive field of 65 kV/cm at a pump-drive delay of $t_{pd} = 38.7$ ps, where two THG contributions are resolvable and labelled by $(3f)_P$ and $(3f)_E$. (b) Integrated THG intensity I_{3f} versus optical pump fluence I_{op} , as derived from the time-domain signals in (a) for the two contributions $(3f)_E$ and $(3f)_P$ separately.

trast, we focus on picosecond ultrafast dynamics in this work.

The resolved distinct differences in the time-domain THG traces demonstrate the powerfulness of our time-resolved high-harmonic generation spectroscopy, which enables us to identify and investigate coexisting dynamical processes from unbound plasma or bound excitonic pairs in the strongly interacting nonthermal many-body system. While the non-parabolic dispersion relation in the conduction and valence bands [18] can lead to the third-harmonic yields of the THz driven charge dynamics, our results show that the driven plasma and excitonic dynamics behave very differently on several important aspects.

Firstly, we study the different nonlinear responses of the THz driven plasma and excitonic dynamics by varying the THz field strength. In Fig. 2(a), we present the results for $t_{pd} = 38.7$ ps, where the excitonic THG traces $(3f)_E$ are behind and clearly separated from the plasma THG $(3f)_P$ in the time domain. With increasing THz drive fluence, the excitonic THG increases much more evidently than the plasma THG. By performing Fourier analysis of the corresponding time-domain traces separately, we obtain plasma and excitonic THG intensity spectra in the frequency domain, from which we derive integrated THG intensity I_{3f} versus THz drive fluence I_f as displayed in a log-log scale in Fig. 2(b). As shown by the solid lines, the excitonic THG exhibits a nearly perturbative cubic power-law dependence of $I_{3f} \propto I_f^{2.9 \pm 0.1}$, whereas the plasma THG follows a linear dependence on the THz drive. The linear dependence of the plasma THG is attributed to the photoinduced change in the THz optical conductivity due to the photoexcitation of charge carriers and

their intraband dynamics, while the nonlinear dependence of excitonic THG arises naturally from band nonparabolicity.

Secondly, with a fixed THz drive field of 65 kV/cm, we study the evolution of the THz THG by changing the optical pump fluence I_{op} (see Fig. 3). We have found distinctly different pump-fluence dependencies for the excitonic and the plasma THG. For $t_{\text{pd}} = 38.7$ ps, a THz third-harmonic yield is hardly resolvable at $I_{\text{op}} = 0.2 \mu\text{J}/\text{cm}^2$ [Fig. 3(a)], but already very evident with a relatively low pump fluence of $I_{\text{op}} = 2.5 \mu\text{J}/\text{cm}^2$. At low fluences, the increase of the excitonic THG yield with enhanced pump fluence is very rapid [see also Fig. 3(b)], whereas the observed plasma THG remains weak. The maximum excitonic THG is reached around $I_{\text{op}} = 12.4 \mu\text{J}/\text{cm}^2$, which is followed by a continuous decrease towards the highest fluences. In contrast, the plasma THG increases monotonically and tends to saturate above about $I_{\text{op}} = 100 \mu\text{J}/\text{cm}^2$.

The integrated THG intensity of the plasma and excitonic dynamics is plotted separately in Fig. 3(b) as a function of the optical pump fluence. The distinct difference in their optical pump dependencies confirms the assignment of the two THG traces as plasma and excitonic THG, respectively. The observed behaviors can be understood as follows. At low pump fluences corresponding to a low charge density, the initial increase of charge density caused by higher pump fluences leads to the formation of more excitons, hence enhancing both the plasma and excitonic third-harmonic yields. At the same time, the interactions between charges and excitons are increasing with charge density. Above a critical charge density, dissociation of excitons occurs due to screening mediated by the enhanced Coulomb interaction. They decay into unbound electron-hole pairs and contribute to the plasma, and eventually excitons cannot be formed in the presence of a large electron-hole density, i.e., in a Mott phase [17, 30]. Therefore, above a critical density a decrease of the excitonic THG is observed experimentally [Fig. 3(b)], while the plasma THG increases monotonically towards saturation.

From the excitation pulse energy corresponding to the strongest excitonic THG observed at $I_{\text{op}} = 12.4 \mu\text{J}/\text{cm}^2$, we can estimate the photoexcited electron-hole pair density as $n = 3.3 \times 10^{16} \text{ cm}^{-3}$, while the excitonic THG essentially vanishes beyond $I_{\text{op}} = 120 \mu\text{J}/\text{cm}^2$ corresponding to $n \gtrsim 3 \times 10^{17} \text{ cm}^{-3}$. Just for comparison, the theoretical Mott density of $n_{\text{Mott}} = 4.4 \times 10^{17} \text{ cm}^{-3}$ at room temperature under thermal equilibrium is not much greater, which is given by assuming extrapolation of the known theoretical results [31, 32] to room temperature. However, due to strong thermal fluctuations the excitons in silicon cannot survive at room temperature under thermal equilibrium conditions. Hence, the observed carrier-density-dependent behaviors here should be understood as transient dynamical properties of the strongly interacting many-body electron-hole system far from thermal equilibrium.

Thirdly, at different pump-drive delays, the overall THz THG intensity exhibits evidently different dependence on the optical pump fluence (see Fig. 4), reflecting the distinct differ-

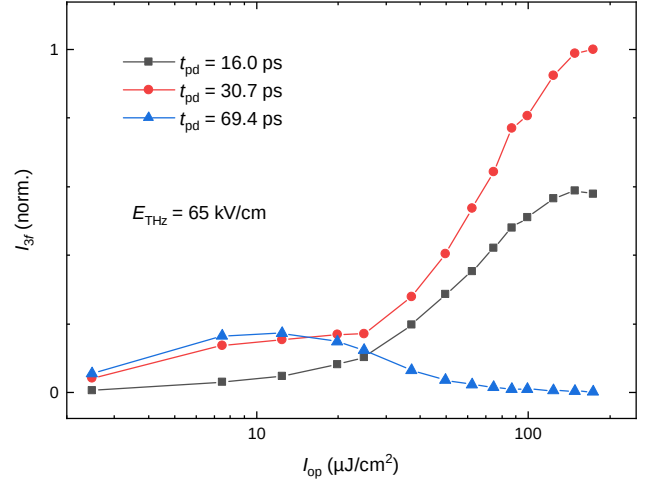


Figure 4. Optical pump fluence dependence of the overall third-harmonic intensity measured with a peak THz electric field of 65 kV/cm for various pump-drive delays.

ence of plasma and excitonic dynamics. For a shorter pump-drive delay, e.g. $t_{\text{pd}} = 16.0$ ps, where a large density of excitons has not been formed, the observed third-harmonic yield is governed by plasma dynamics, therefore a monotonic increase of THG due to increased charge-carrier density at a higher pump fluence is observed [see Fig. 1(c)-(e) and Fig. 4]. In contrast, at a rather long pump-drive delay of $t_{\text{pd}} = 69.4$ ps, where the optical pump pulse is well ahead of the THz drive pulse [see Fig. 1(a)], the observed third-harmonic yield is mainly due to the driven excitonic dynamics [see Fig. 1(f)], leading to a Mott-type nonmonotonic dependence of THG intensity (Fig. 4). At low fluence the THG intensity increases due to larger density of excitons, but above the critical value of $12.4 \mu\text{J}/\text{cm}^2$ the third-harmonic yield decreases monotonically due to the suppression of the picosecond exciton formation by enhanced Coulomb interactions [c.f. the $t_{\text{pd}} = 69.4$ ps curves in Fig. 1(c)-(e)]. For an intermediate pump-drive delay (see $t_{\text{pd}} = 30.7$ ps in Fig. 4), the third-harmonic yield at low pump fluences is characterized by excitonic dynamics as featured by a broad maximum, whereas at high pump fluences dominated by plasma contributions.

In conclusion, by using time-resolved optical pump THz third-harmonic generation spectroscopy, we have investigated picosecond nonthermal nonlinear many-body dynamics of an optically excited strongly interacting electron-hole system in high-purity silicon at room temperature. We have shown that under the drive of an intense narrowband THz pulse, both the plasma and excitonic dynamics lead to the emission of THz radiation at the third-harmonic frequency. For plasma, this corresponds to a broadband radiation dominated by a linear THz field dependence, which is dominant when the pump and drive pulses overlap. In contrast, the driven excitonic third-harmonic emission is narrowband and characterized by a nonperturbative power-law THz field dependence. Moreover,

with increasing charge-carrier density the plasma emission increases monotonically, while the excitonic third-harmonic generation exhibits a maximum at a critical density, above which strong Coulomb interactions lead to dissociation of the excitons, featuring a Mott crossover behavior.

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